

Final Research Output

MealCare: Food Waste at UofT

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Brief Overview

This MealCare project examines post-consumption food waste in the University of Toronto St. George campus dining halls through student surveys, structured waste audits, and interviews with the dining staff. Surveys revealed key waste drivers like poor taste, undercooking, and fluctuating appetites rather than oversized portions with students prioritizing preventative solutions like flexible serving options, and pre-portion taste test. Audits conducted across multiple dining halls during peak times confirm consistent waste patterns, while staff insights exposed gaps in training, protocols, and communication around waste reduction practices. These findings highlight the need for more targeted interventions that bridge student behaviors and institutions practices to foster sustainable dining habits on campus.

Survey

We surveyed 154 students from the UTSG campus, and the top 3 categories being thrown away are carbohydrates (48.1%), followed by vegetables and fruits (14.9%), and then proteins (13%), which perfectly aligns with our audit data (see Appendix B). Additionally, the collected data suggested that oversized portions are not the primary reason people discard food at dining halls. In fact, over 50% of people who reported inappropriate portion sizes actually thought they were too small. To add, there is no direct causal relationship between portion sizes and total food waste. According to the survey, students discard food for other reasons that are unrelated to quantity - such as poor taste, undercooking, stressed eating, or, just simply a lack of appetite. In other words, food waste appears to be occasional rather than habitual.

To address these qualitative drivers, respondents identified three primary intervention strategies for waste mitigation. 37.7% of the participants suggest implementation of flexible

portion options, allowing for the "right-sizing" of meals to match fluctuating appetites. This is followed by the provision of taste tests prior to committing to a full portion (24.7%), a strategy specifically designed to reduce waste resulting from unpalatable or undercooked food. Finally, 16.9% of respondents highlighted composting uneaten food as a critical secondary measure for recovering unavoidable waste. Collectively, these results suggest that students seem to favor preventive strategies (i.e., reduce waste before it happens) rather than relying only on disposal methods. For more detailed information regarding survey results, please see the attached link in Appendix B.

Audit

Method

The study employed a structured food waste audit conducted in four dining halls, including Canada Room, Starchan, Audrey Taylor, and Howard Ferguson, over four-hour periods covering peak meal times. Data collection consisted of four components. First, plate counts were recorded by positioning the researcher near disposal areas to document the total number of discarded plates and the number containing visible edible leftovers, excluding inedible components. Second, a visual assessment was conducted to identify commonly discarded food categories, with the most frequently wasted item recorded during each audit period. Third, subsample weighing involved approaching at least three consenting diners after they finished eating, photographing the remaining food, and measuring its weight using a digital scale, along with the weight of the empty container. Participants were informed of the research purpose before data collection. Fourth, participant observation was used to document qualitative insights on dining behaviours, including brief notes on patterns, interactions, and notable events. All observations were recorded in an audit

diary. Quantitative data included plate counts and weight measurements, while qualitative data consisted of observational notes. Data were recorded immediately during each audit session to ensure accuracy and consistency across observations. Standardized procedures were followed to maintain reliability and minimize variability across audit sessions.

Findings

The audit's findings show that food waste is a consistent and consequential issue across all four dining halls. It was observed that between 37% and 39% of the plates contained edible leftovers, and more than one-third of the meals were not being fully consumed. Collectively, the audit period produced an estimated 94.2 kg of food waste, with an average of approximately 150.16 grams wasted per plate. These patterns were consistent in both the weekday and weekend observations, further demonstrating that food waste is a widespread and continuous issue across dining halls.

Interview

Method

This study employed a qualitative interview approach to explore post-consumption food waste and student dining patterns at the University of Toronto St. George (UTSG) dining halls. Semi-structured interviews were conducted with dining hall staff to gain insight into food handling practices, institutional policies, and observed student behaviors. Participants were recruited on-site at selected dining halls where the audit was done, based on availability and willingness to participate.

Each interview lasted approximately 10–15 minutes and followed a structured interview guide consisting of warm-up, core, and probing questions. Interviews were audio-recorded with participant consent to ensure accuracy in data collection. Ethical considerations were prioritized: participants were informed of the study's purpose, assured of confidentiality and anonymity, and reminded that participation was voluntary, with the option to skip questions or withdraw at any time.

The interview questions focused on three key areas: dining hall policies and practices regarding leftover food, staff training and waste reduction strategies, and staff observations of student dining patterns and food waste trends. Probing questions were used to elicit more detailed responses when necessary. Audio recordings were later reviewed and summarized to identify recurring themes and patterns relevant to food waste procedures in the dining halls.

Findings

Two interviews were successfully conducted with staff members at Howard Ferguson (University College) and Audrey Taylor (New College). Other dining halls were less cooperative in participating in interviews. At Audrey Taylor, the staff member could not answer any specific questions about waste reduction protocols at the pre-consumer level at the dining hall. They explained that small amounts of food waste are composted and large amounts are donated, but could not elaborate on where they are donated to, or what constituted a large enough amount to be donated, and suggested it was up to the discretion of the chef. The interviewee at Howard Ferguson had similar answers; although this individual was not a kitchen staff member, they explained they had not received any waste reduction-related training for their position. This staff member explained that student surveys are conducted by the dining hall to gauge student interest in dishes,

citing meat pies as a successful dish example. This allows them to plan meals and order ingredients accordingly, as well as rotate their dishes and meet student demand.

Our goal was to compare interview answers of kitchen staff and dining hall managers to investigate if staff members knew about and were conducting the alleged protocols described by managers, but ultimately, the managers were difficult to contact and were hesitant to participate for the most part. There were also communication issues between the two; when we reached out to managers to get permission to conduct interviews in the dining halls, it was not communicated to on-duty staff, who then declined to be interviewed without hearing from managers.

Recommendations for MealCare

Our study found that there are disparities between what each campus dining hall is doing to prevent food waste, and that it remains a somewhat problematic volume of waste. Our recommendations for how MealCare can remedy this are to 1) target students to manage their contributions to the problem, 2) increase awareness of preexisting solutions, and 3) work together with the University of Toronto's dining halls to build solutions to fit their needs.

The first recommendation to target students emerges from the behaviours we witnessed while conducting the audits. At the all-you-can-eat buffet-style dining halls, students often took too much food or would discard one dish and later go back for more of another kind. We believe that one way to target the issue is to make students more aware of this behaviour, through posterage or Instagram posts. Eating can become a monotonous task, especially for students who are focused on their studies. We even observed students studying as they ate, meaning things such as food waste are not an issue that crosses their minds. A simple reminder next to the plates or trash cans could persuade students to think twice about how much they are taking. We have provided a

sample poster, but they could take many forms, potentially increasing in effectiveness if they take on a more funny or attention-grabbing tone. As for the Instagram post, it should aim to more seriously inform students about the issue of waste on campus in the hopes they take action of their own volition.

The second recommendation is to increase awareness and hopefully usage of MealCare's existing projects, such as the campus community fridges. Our team members had never heard of this until researching MealCare, and we believe that more students would make use of the fridges if they knew about them. This can also be remedied by posterizing and Instagram posts, but also through the various email newsletters at the school, or by asking professors to make announcements to their classes, primarily targeting first-year students through these avenues, as they make up most of the dining hall users.

Our third recommendation is to get involved with the dining halls to facilitate more specific solutions. This would ultimately be shaped in collaboration with the dining hall managers, but to start, we recommend implementing a method from the University College dining hall, Howard Ferguson, at the other dining halls, which is student surveys. These surveys are for estimating student demand for specific dishes. This would be a helpful waste reduction method, because, as we found in our survey, disliking the dishes was a leading cause for waste. New College mentioned that they donated food that was not eaten by students. MealCare could also help facilitate this for other dining halls and help build community connections with local shelters or meal aid programs. Finally, waste audits could be conducted more regularly to gauge the effectiveness of the implementation of our recommended methods. Getting dining halls and students involved in the audits by setting achievable waste targets, or even getting dining halls to participate in a friendly

competition for who can reduce their waste the most, would be a good way of boosting morale and engagement.

We understand that MealCare is ultimately limited by resources and volunteers, so these recommendations aim to give a range of possibilities, and could perhaps be aided by future students of ENV421. Given the results of our waste audit, staff interviews, and student surveys, we recommend the following upstream, downstream, and system-level strategies.

Upstream Interventions

These strategies focus on influencing student behaviour before the waste occurs, addressing root causes of post-consumer food waste, such as decision making, awareness, and habits.

Expansion of Awareness Campaigns

- **Dining Hall Posters:** Listing food waste statistics, or catchy messaging like “Take what you'll eat, eat what you take” to compel consumer behaviour
- **Rebooting the MealCare UofT Instagram account:** to spread awareness about food waste on campus & to promote MealCare’s resources.
- Awareness via school newsletter emails & course announcements

Downstream Interventions

Expand Food Redistribution

- Increasing awareness & access to community fridges via Instagram & posters
- Implement post-consumer food waste solutions, like on-campus composting. See the following link for an example of what the UofT Scarborough campus is doing with their vermicompost system: <https://www.utsc.utoronto.ca/food/vermicompost-campus>

System-Level Interventions

Collaborating with dining halls to implement findings from the waste audit & survey of dining hall students. Our findings suggest that much of the food waste occurs because of qualitative and behavioural factors, such as food quality & fluctuating appetite, rather than portion size alone.

Suggestions:

- For pre-portioned meals, consider smaller carbohydrate portions since this is the most discarded food item (at 48.1%)
- Implementing strategies that appealed to students from the survey, such as a “try before you buy” program
- Regular monitoring, waste audits, and surveys to help inform dining hall staff of what is actually wanted by the community to produce waste.
- Collaborating with other organizations across campus to create a more circular economy on campus. For example, composting initiatives for post-consumer waste, which could be reused on campus in our greenhouses.

Recommendations for Continuing Research

Expansion of research questions & methodology to better target and represent the actual waste behaviours of the community.

Expansion of Research Question

The current waste audit methodology focuses primarily on quantifying post-consumer food waste produced on plates in the dining halls. However, after observing purchasing patterns of students in the dining hall, we noticed a large proportion of the food bought was pre-packaged.

Thus, the current research question and methodology fail to capture the full scope of food waste on campus, thereby limiting the comprehensiveness and representativeness of the findings within the actual community. Furthermore, we suggest extending this research to encompass more dining halls across campus, both public and private dining halls (such as Oak House Residence), as well as to other on campus food halls & cafes, such as Robarts, Sid and MSB Food Hall, as well as Cafe Reznikoff and The Exchange. Expanding the research to all these different sectors will provide a more comprehensive understanding of the problem of food waste on the University of Toronto St. George campus.

Audit Methodology Refinements

Working off the waste methodology of last year's research, we suggest the following suggestions to tweak (1) accuracy, (2) the scope of the research question.

1. In our experience, the use of food scales to record data produced a margin of error, as empty plates did not consistently measure the same weight, despite being identical plates. This suggests a potential measurement error, since readings varied based on the plate placement on the scale, indicating sensitivity and potential calibration limitations. Also, these measurements were randomly sampled, producing an estimate of the total food waste, rather than directly measuring it. Alternatively, we suggest partnering more collaboratively with the dining halls during these audits, and instead having a separate compost bin where students can dump their food waste, for the entire bag to be weighed post audit. This removes the measurement error from the scales and provides a more precise measure of the food waste, as opposed to an estimate from a smaller subsample.
2. The current methodology only records food wasted on plates. A small change to the methodology could estimate the proportion of students who are consuming pre-packaged

items. By recording both the total number of sales (plates+prepackaged) and measuring only the plates returned (as performed in the current methodology), then subtracting plates from the total purchases, one can demonstrate the proportion of students consuming to-go food items. This will give a more accurate read of what the actual behavioural and purchasing pattern of the students is, and inform whether the research question should expand to these pre-packaged food items, to address both food & packaging waste.

Advice for future studies

Communication with the dining halls was much more difficult than we had anticipated. We recommend getting in touch with the dining halls as soon as possible during the future research process. Permission will be required to put up posters & to enter into select dining halls, so having this figured out as early as possible will streamline the research process. We have included the contacts below of all the dining halls we intended to visit (see Appendix C). Note, we only received access to & data from the following dining halls: New College, Trinity, University and St. Michael's College. We recommend that future studies expand the suit to more dining halls across campus, including Victoria college and oak house, which we did not get to sample.

Conclusion

This project highlights food waste as a persistent challenge rooted in behavioral habits and operational inconsistencies that urge proactive strategy over disposal methods. Mealcare can drive change through raising student awareness through poster or Instagram campaigns with messages like “Take what you’ll eat,” and the promotion of existing resources such as the campus community fridges. Collaboration with the dining halls, through student demand

surveys, facilitated donations to local programs, regular audits and even friendly waste reduction competitions offer practical and scalable solutions. These steps foster individual accountability while stimulating progressing institutional momentum for sustainability, and inspiring border campus-wide changes and future community efforts. By implementing these measures, MealCare can achieve waste reduction over time, while building a model for other universities facing similar issues. Ultimately, empowering students and staff as active panthers in the process cultivated an environment of stewardship that can extend beyond dining halls to everyday campus life.

Appendix

Appendix A: Audit Data for Weekdays and Weekends

Table 1. Weekend Audit Data (2026-02-22)

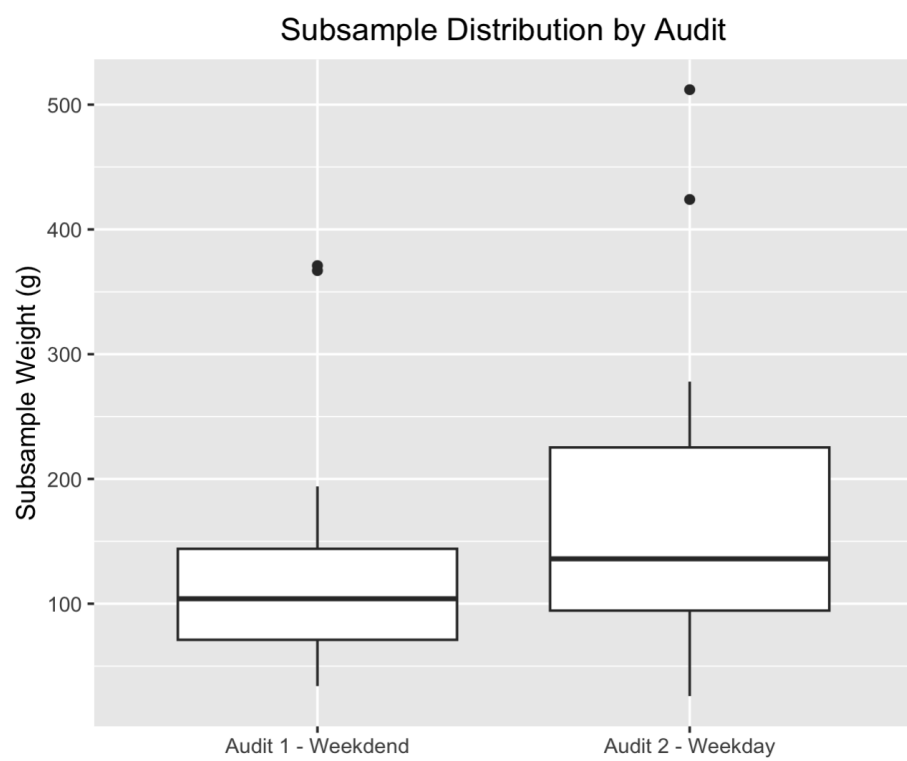
Dining Halls	Discarded Plates with Edible Food	Total Plate Count	Proportion of Leftover Plates
Starchan	27	95	28.4%
Howard Ferguson	55	125	44%
Canada Room	47	124	37.9%
Audrey Taylor	81	245	33.1%
Total	210	590	35.6%

Table 2. Weekday Audit Data (2026-02-23)

Dining Halls	Discarded Plates with Edible Food	Total Plate Count	Proportion of Leftover Plates
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Starchan	47	136	34.6%
Howard Ferguson	55	125	44%
Canada Room	117	300	39%
Audrey Taylor	198	507	39.1%
Total	417	1,068	39%

Figure 1. Subsample Distribution by Audit (Weekend versus Weekday)



Appendix B: Descriptive Statistics for Survey Results

Link to Survey Summary:

https://forms.office.com/Pages/AnalysisPage.aspx?AnalyzerToken=mSz49Yb3COtDp4EuDQvO0j6Z6sNSnzbe&id=DQSIkWdsW0yxEjajBLZtrQAAAAAAAAAAAAAO__SOMACVURDQ4RUpJUDdMTTQxUUkzT1BVMDJPWE9NVC4u

Appendix C: Dining Hall Contact Information

DINING HALL	CONTACT	ADDITIONAL INFO
Victoria College (Burnwash Dining Hall)	Jason Sabatini jason.sabatini@utoronto.ca	89 Charles St W https://www.vicu.utoronto.ca/hospitality-services/student-meal-plans-and-dining-hall-menus/burwash-dining-hall-and-neds-cafe -Monday to Friday 7:30 am ~ 7:30 pm -Weekends & Holidays 10:00 am ~ 2:30 pm, 4:00 pm ~ 7:30 pm
Oak House Residence (Oak House Dining Commons)	oakhouse.residence@utoroto.ca	700 Spadina Ave https://foodservices.utoronto.ca/oak-house-dining-commons/ -Monday to Thursday 7:00 am ~ 9:00 pm -Friday 7:00 am ~ 8:00 pm -Saturday and Sunday 8:00 am ~ 8:00 pm

New College (New College Dining Commons)	Residence spaces & experience new.residence@utoronto.ca	40 Willcocks https://foodservices.utoronto.ca/new-college-dining-commons/ -Monday to Thursday 7:00 am ~ 10:00 pm -Friday 7:00 am ~ 9:00 pm -Saturday 8:00 am ~ 9:00 pm -Sunday 8:00 am ~ 10:00 pm
Trinity College (Strachan Hall)	trinity.deanofstudents@utoronto.ca	6 Hoskin Avenue https://www.trinity.utoronto.ca/engage/dining-at-trinity/ -Monday to Friday 7:30 am ~ 8:00 pm -Weekends & Holidays 8:30 am ~ 8:00 pm
St.Micheals College (Canada Room)	smc.residence@utoronto.ca https://stmikes.utoronto.ca/about-us/contact-us	81 St. Mary Street https://stmikes.utoronto.ca/community/st-michaels-college-residence/residence-

		experience/residence-life-meal-plans -Monday to Friday 7:30 am ~ 9:30 pm -Weekends & Holidays 7:30 am ~ 8:30 pm
University College (Howard Ferguson Dining Hall)	uc.food@utoronto.ca	75 St. George St https://www.uc.utoronto.ca/students-current-students-food-howard-ferguson-dining-hall -Monday to Friday 7:30 am ~ 7:30 pm (closed between 3 to 4 pm) -Weekends 9:00 am ~ 7:30 pm (closed between 3 to 4 pm)

Instagram Post

An awareness-based intervention was designed using an Instagram campaign to promote more sustainable student dining patterns and reduce post-consumption food waste at UTSG dining halls. Instagram was selected due to its popularity among university students and its effectiveness in delivering visually engaging and easily digestible content. The suggestible campaign centers on

simple, behavior-focused messages such as “Take what you’ll eat, eat what you take,” encouraging students to be more mindful of portion sizes and food choices.

The posts incorporate clear visuals and concise captions to communicate actionable steps, including starting with smaller portions, selecting preferred foods, and finishing meals before discarding waste. The design prioritizes high-contrast visuals, minimal text, and relatable dining hall scenarios to capture attention and enhance message retention.

To ensure effectiveness, several guidelines were followed: messages were kept simple and focused, content was tailored to student behaviors, and recommendations were practical and easy to implement. Additionally, the campaign emphasizes small individual actions while highlighting their collective environmental impact to encourage behavior change.

Overall, this intervention aims to influence student habits at the point of consumption, complementing institutional efforts to reduce food waste and promote sustainability on campus.



Figure 1: Example Instagram Post



**Small changes can
reduce waste:**

- ✓ Start with smaller portion sizes
- ✓ Choose foods you actually want
- ✓ Go back for more if needed
- ✓ Avoid taking food "just in case"

Dishing Facts:

~ **37.3%**

of plates had
leftovers

~ **94 kg**

of **food waste**
is produced in
just 8 hours*

*data across 4 UofT dining halls

**TAKE WHAT YOU'LL EAT.
EAT WHAT YOU TAKE.**

Figure 2: Example Instagram Post/Dining Hall Poster

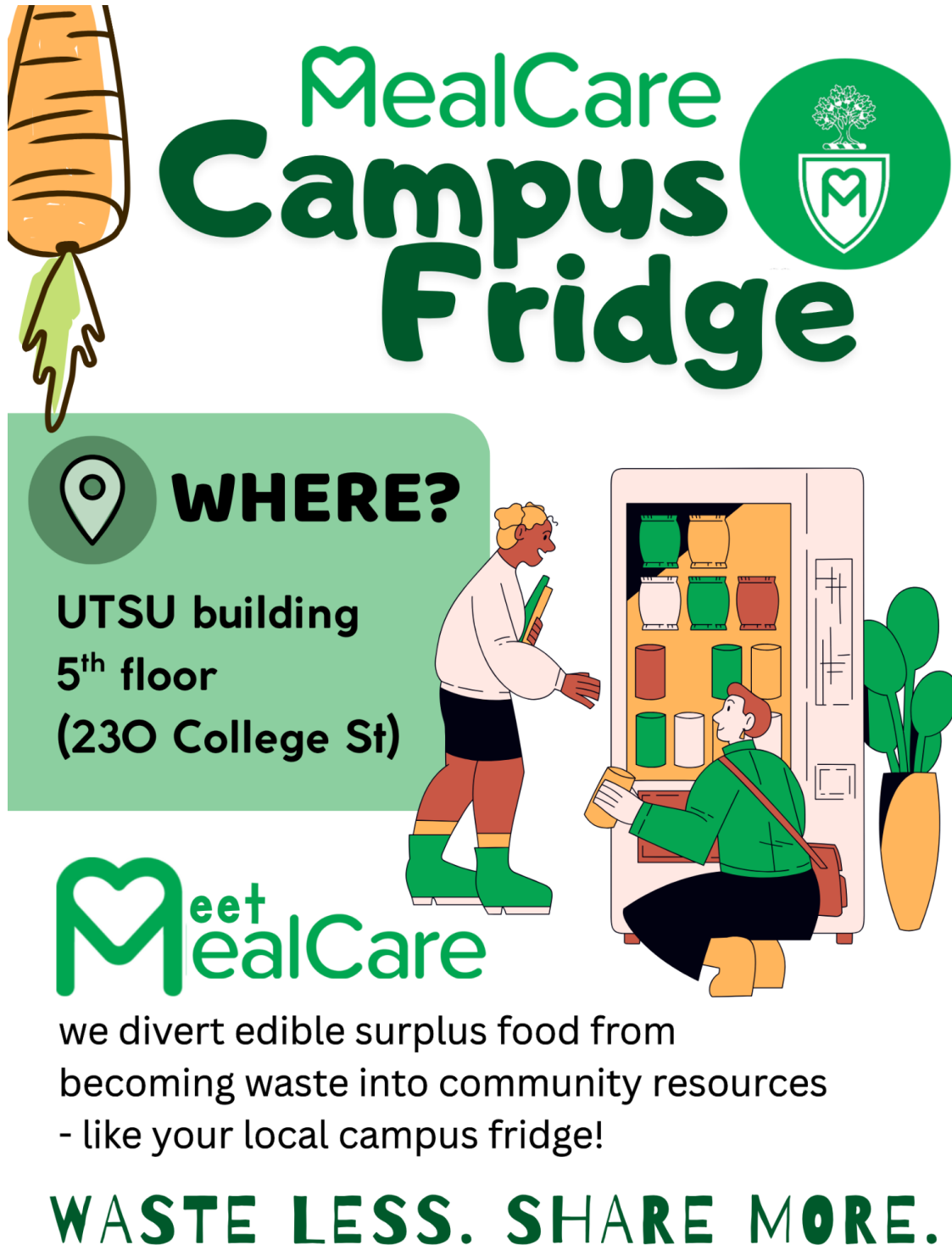


Figure 3: Instagram post & Campus poster